Reticulation-Aware Network Comparison in PhyNetPy

A validation and comparison report for the phylogenetic-network dissimilarity measures in PhyNetPy, centred on the new reticulation-aware, tripartition-matching measure of Nakhleh (2026).

Contents

All numbers in this report are computed live by scripts/reticulation_distance_report.py against the installed phynetpy package; regenerate with:

```
python scripts/reticulation_distance_report.py
```

1. Design & Integration

The measure lives in phynetpy.ReticulationComparison and is
re-exported from phynetpy.GraphUtils and the top-level package.

```
SINGLE ENTRY POINT
    from phynetpy import compare_networks
    result = compare_networks(reference, inferred)
    D, D_ret_hat, prec, rec = result   # iterable => (D, D_ret, P, R)

The result is a NetworkComparison dataclass exposing:
    .D             global Nakhleh metric (Eq. 5, unnormalized)
    .D_hat         bounded global score 2D/(|V1|+|V2|) in [0,1]
    .D_ret         reticulation dissimilarity (Eq. 4, metric)
    .D_ret_hat     normalized reticulation dissimilarity in [0,1]
    .precision/.recall/.f1   reticulation-level recovery (Def. 4)
    .matching      the optimal (ref_index, inf_index, cost) pairing
    .combined(lam)             D_lambda = lam*D_ret + (1-lam)*D

PARAMETERS (all optional, with the note's defaults)
    distance = 'jaccard' | 'symmetric'   ground set distance
    rho      = deletion penalty for an unmatched reticulation
               (default = maximally-wrong = 3 under Jaccard)
    tolerance= tau; a matched pair counts as 'recovered' iff its
               cost <= tau (tau=0 demands exact recovery)
    alpha    = block weight; alpha=1 is the leaf-set measure,
               alpha<1 adds the topology-aware refinement (Sec. 8)

CONVENIENCE WRAPPERS
    reticulation_dissimilarity(n1, n2, normalize=True) -> float
    reticulation_precision_recall(ref, inf) -> (prec, rec, f1)
    nakhleh_metric(n1, n2) -> D            (also normalize=True)
    combined_dissimilarity(n1, n2, lam=0.5) -> D_lambda
    reticulation_tripartitions(net) -> [ReticulationTripartition]

WHY IT EXISTS
    Reticulation is rare relative to the tree-like bulk of a network,
    so a global measure can report near-perfect agreement while every
    inferred hybridization is wrong (the base-rate fallacy).  D_ret
    scores ONLY the reticulation content; it is a pseudometric on
    networks (blind to tree-only edits) and pairs with the global D
    via D_lambda to recover a genuine metric.
```

2. Catalogue of Available Dissimilarity Measures

measure	what it scores	range	properties
D_ret (norm)	reticulation events only	[0,1]	pseudometric; NEW
D_ret (raw)	reticulation events only	[0,inf)	metric on signatures; NEW
D (Nakhleh)	whole network (subnetworks)	[0,inf)	metric (reduced nets); NEW
D_lambda	convex mix of D_ret & D	[0,inf)	metric for lambda<1; NEW
mu-distance	path-multiplicity vectors	[0,inf)	metric (tree-child)
tripartition	edge tripartitions	[0,inf)	NOT a metric
hardwired	hardwired clusters	[0,inf)	dissimilarity
softwired	softwired clusters	[0,inf)	dissimilarity
displayed-tree	displayed tree set	[0,inf)	dissimilarity
Robinson-Foulds	non-trivial clusters	[0,inf)	metric (trees)

Only the reticulation-aware family (D_ret, D, D_lambda) isolates the hybridization signal. Every other measure aggregates over the whole network and is therefore dominated by its tree-like structure -- the central concern the new measure addresses. 'tripartition' here is the classical PhyloNet edge-tripartition dissimilarity, which is provably unable to separate some non-isomorphic networks (see Section 4).

3. Boundary Sanity Checks (Section 6.1 / E5)

scenario	Dret_n	Dret	D	Dl.5	mu	trip	hard	soft	disp	RF	Prec	Rec
two distinct trees	0	0	2	1	2	2	2	2	2	2	1	1
network vs its underlying tree*	1	9	7	8	14	10	7	9	7	7	0	1
self-comparison (N vs N)	0	0	0	0	0	0	0	0	0	0	1	1

Expected (Section 6.1 / E5): two trees agree perfectly on reticulate content (D_ret=0) while the global measures still separate them; measuring a network against a tree gives D_ret=1 with precision 0 (every reticulation is a false positive); self-comparison is 0 throughout with precision=recall=1.
* here 'Tree balanced' stands in as an underlying tree of N; only the reticulation columns and Prec/Rec are meaningful for that row.
Headers: Dret_n = normalized D_ret, Dret = raw D_ret, D = Nakhleh metric, Dl.5 = D_lambda(0.5); mu/trip/hard/soft/disp/RF per catalogue.

4. Literature Stress Test (Section 11)

Cardona, Rossello & Valiente exhibit explicit non-isomorphic tree-child networks that the classical tripartition metric cannot tell apart. We reconstructed the pair exactly from the mu-vectors in Tables 1 & 3 of arXiv:0708.3499 (their Figs. 4 and 8). Both are networks on {1,2,3,4,5} with three reticulations A, B, C of clusters {2,3,4}, {3,4}, {4}; the two networks differ only by swapping which hybrid (B or C) hangs beneath internal nodes b and c.

Reticulation tripartitions of N (identical multiset for N'):

```
    hybrid A:  A=['5']  B(cluster)=['2', '3', '4']  C=['1']   [proper]
    hybrid B:  A=['2']  B(cluster)=['3', '4']  C=['2', '5']   [IMPROPER (A∩C≠∅)]
    hybrid C:  A=['3']  B(cluster)=['4']  C=['2', '3', '5']   [IMPROPER (A∩C≠∅)]
```

This confirms Proposition 1 / Remark 1 of the note on a real published network: hybrid A has incomparable parents (clusters {1,2,3,4} and {2,3,4,5}) and a PROPER tripartition, whereas hybrids B and C each have one parent ancestral to the other -- a triangle -- and are IMPROPER, exactly the weak-time-consistency violation the note flags.

Because the two networks share their entire (L(h), reticulation scenario) signature (Lemma 1), D_ret = 0 -- the designed pseudometric behaviour -- while the global metrics D and mu-distance still separate them. Numbers on the next page.

```
    D_ret(N, N')        = 0.000
    mu-distance(N, N')  = 2   (matches the paper)
    tripartition(N, N') = 0   (the documented failure)
    D (Nakhleh)(N, N')  = 3.000   (separates them)
```

4. Literature Stress Test -- the two networks

N (Fig. 4)

```
                                                    r
                                                   / \
                                                  a  b
                                                 / \/ \
                                             1 c
                                              / \
                                      d
                                     / \
                     5                                 A
                                                       |
                                                       e
                                                      / \
                                                   2        B
                                                            |
                                                            f
                                                           / \
                                                         C     3
                                                         |
                                                         4
```

N' (Fig. 8) -- hybrids B and C swapped beneath b and c

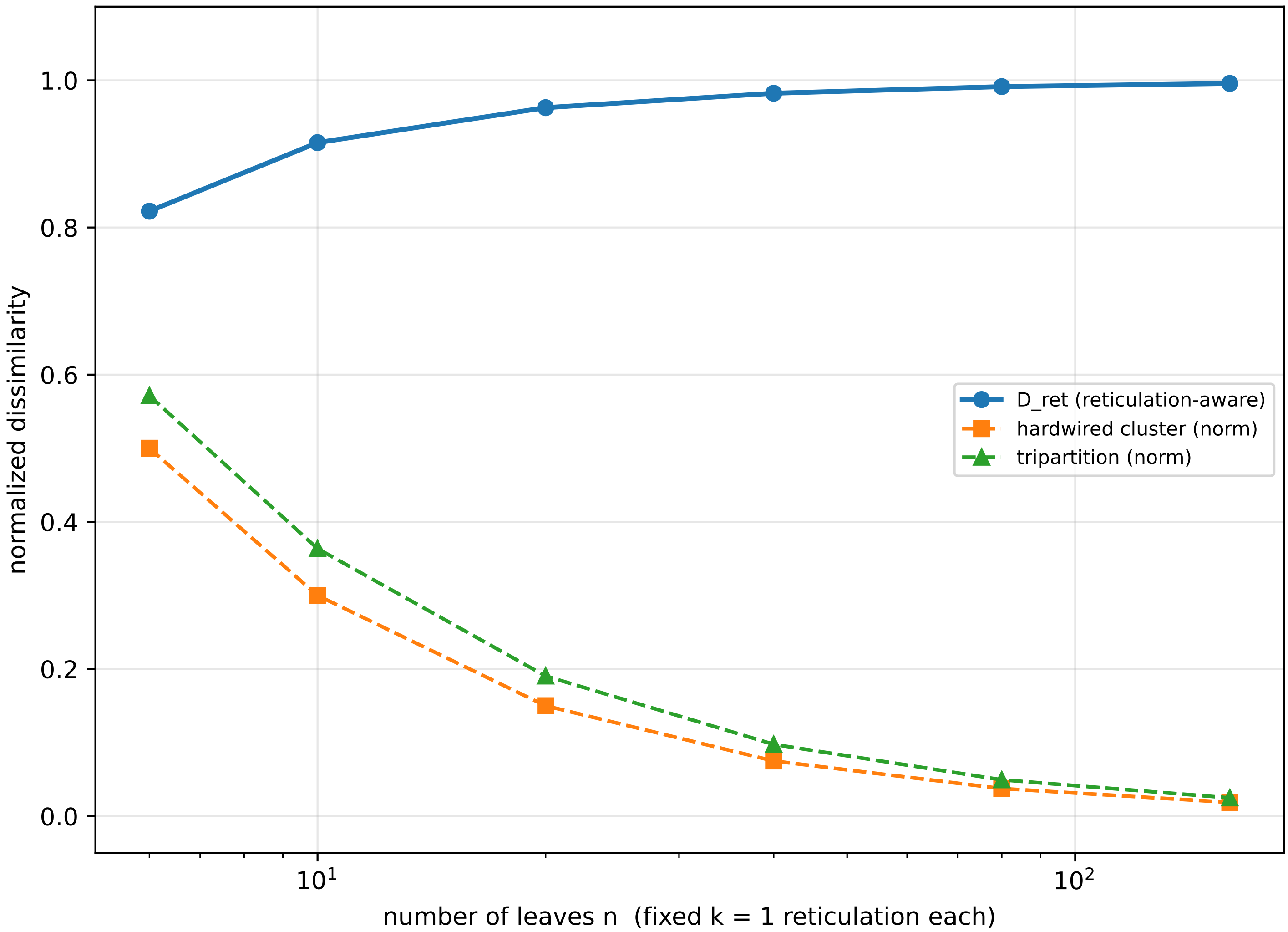
```
                                       r
                                      / \
                      a                                 b
                     / \                               / \
   1                                              c
                                                 / \
                                           d
                                          / \
                                          A  5
                                          |
                                          e
                                         / \
                           2                             B
                                                         |
                                                         f
                                                        / \
                                                   3           C
                                                               |
                                                               4
```

4. Literature Stress Test -- all measures

pair	Dret_n	Dret	D	Dl.5	mu	trip	hard	soft	disp	RF	Prec	Rec	F1
N vs N' (signature-identical)	0	0	3	1.500	2	0	0	2	4	0	1	1	1
N vs N-rewired (1 retic edge)	0.111	1	3	2	4	3	1	4	8	1	0.667	0.667	0.667

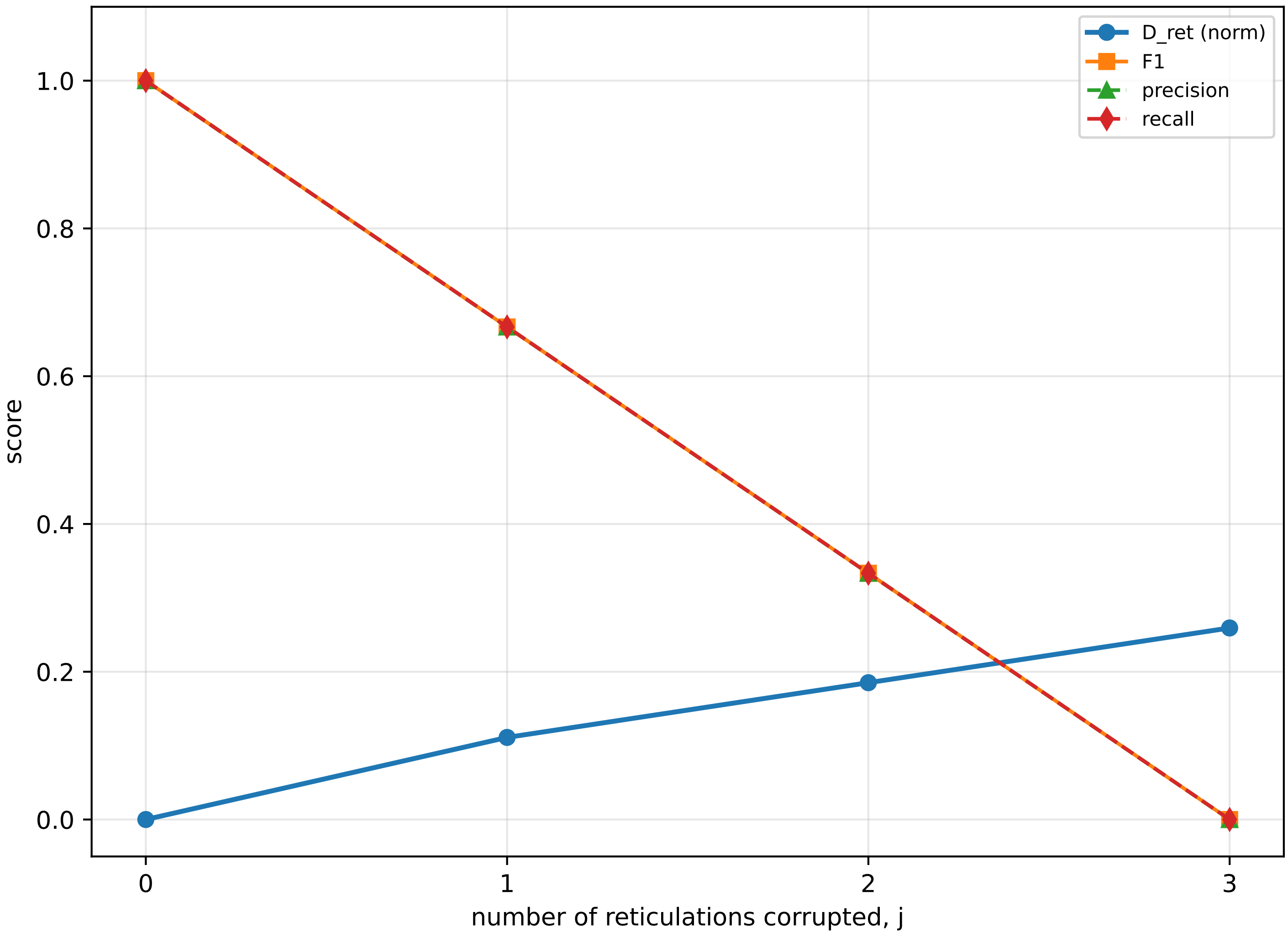
Top row: the adversarial pair. D_ret and the classical tripartition measure both read 0 (identical reticulate signature), but mu-distance = 2 and D > 0 correctly certify the networks are different.
Bottom row: rewiring a single reticulation edge of N perturbs exactly one of the three hybrids, so two of three are still recovered at tolerance 0 -- precision = recall = F1 = 2/3 -- and D_ret jumps off 0, reproducing the (2/3, 2/3, 2/3) worked example of Section 11 (the exact D_ret magnitude depends on the specific edge moved).
Headers: Dret_n = normalized D_ret, D = Nakhleh metric, Dl.5 = D_lambda at lambda 0.5, mu/trip/hard/soft/disp/RF as in the catalogue.

5. Base-Rate Masking (Proposition 4 / E2)



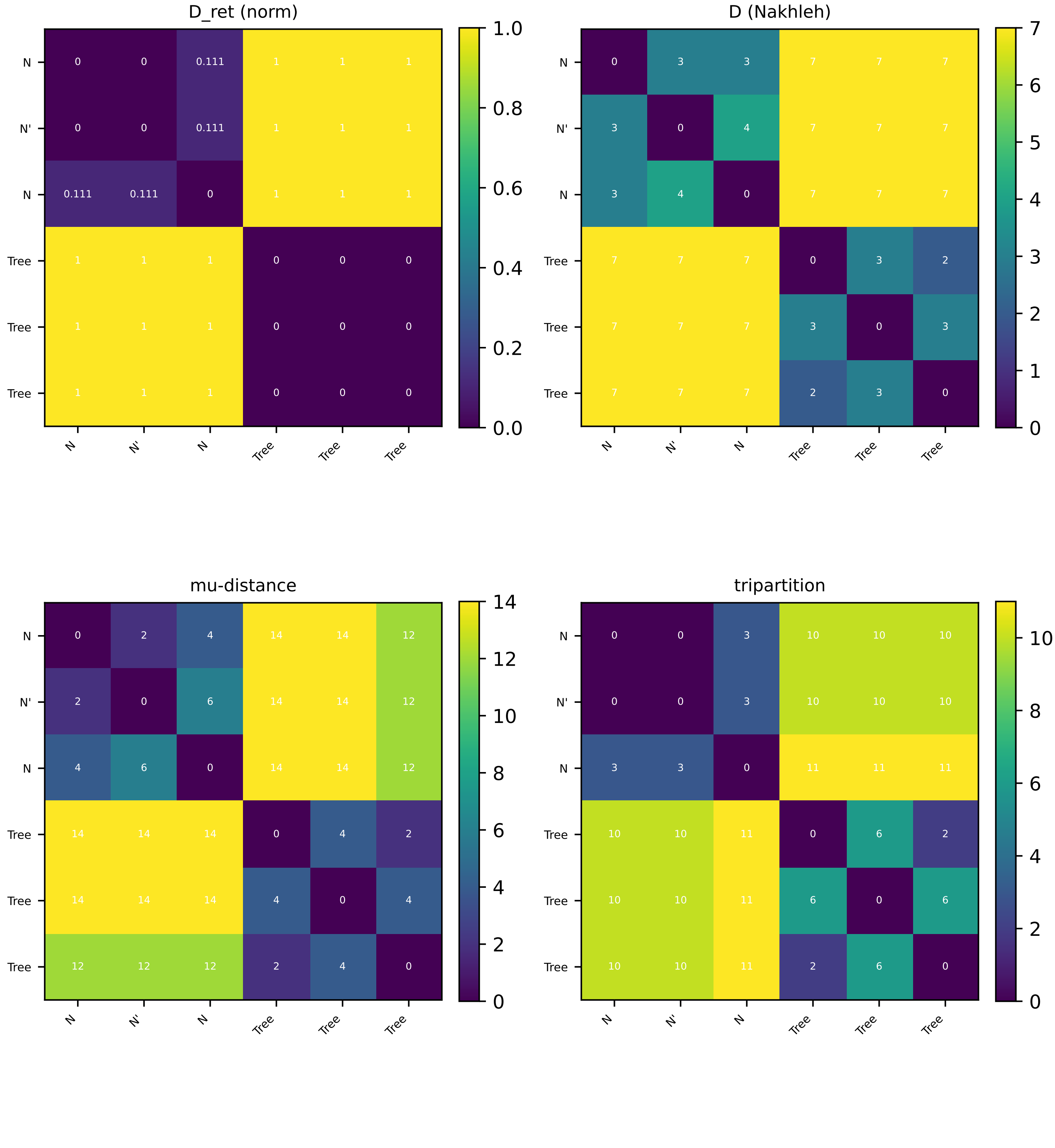
A caterpillar scaffold shared by both networks, with one reticulation placed among the prefix leaves in N1 and among the suffix leaves in N2, so the two hybrid clusters are leaf-disjoint. As n grows the reticulation is diluted against an ever larger tree: the normalized GLOBAL cluster measures decay toward 0 (declaring the networks 'nearly identical'), yet the reticulation-aware D_ret stays near 1 -- it sees that the single event is entirely wrong. This is the base-rate fallacy the measure is designed to expose. (The raw mu-distance grows without bound here and so is omitted from this normalized view.)

6. Monotone Degradation & Recovery (E1)



Starting from N (reference) we corrupt j = 0,1,2,3 of its three reticulations by relocating a hybrid's parent arc to a spurious donor. As j rises the reticulation dissimilarity climbs monotonically from 0 toward 1 while recall / F1 fall from 1 to 0 -- evidence the score is calibrated to the fraction of events recovered. (j=0 is the self-comparison: D_ret=0, precision=recall=1.)

7. Pairwise Comparison of Key Measures



D_ret gives distance 0 for the N/N' pair (shared reticulate signature) and for all tree-tree pairs (no reticulations), while D and mu-distance separate every non-isomorphic pair.